The ALLADER Model: An Attentive Additive Gaussian Process for Ladder-Shaped Data

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1 Model Framework

The proposed ALLADER (Attentive Additive Ladder) model follows the standard Gaussian Process (GP) framework. Given a set of training data, we model the response vector $\boldsymbol{Y}$ as a draw from a Multivariate Normal (MVN) distribution:

$$\boldsymbol{Y} \sim \mathrm{MVN}(\boldsymbol{\mu}, \sigma^2 \boldsymbol{\Psi})$$

where:

- $\boldsymbol{\mu}$: The mean vector, often assumed to be a constant scalar μ.
- σ^2: The **overall process variance**, a scalar parameter that captures the total variability of the output.
- $\boldsymbol{\Psi}$: The **correlation matrix**, a positive semi-definite matrix with ones on the diagonal ($\Psi_{ii} = 1$). It describes the structural relationship between any two data points based on their inputs.

The primary contribution of this work lies in the novel design of the correlation matrix $\boldsymbol{\Psi}$, which is tailored for multi-stage, variable-dimension (i.e., ladder-shaped) input data.

2 The ALLADER Correlation Structure

Inspired by the additive structure of the ‘aLadderGP‘ framework, we design a correlation function $\Psi_{st} = \mathrm{Corr}(Y_s, Y_t)$ composed of weighted "main effect" correlations and "interaction effect" correlations. This construction ensures that Ψ is a valid correlation matrix.

2.1 Step 1: Base Correlation Functions

First, we define the fundamental building blocks of the correlation structure.

- **Stage-wise Main Effect Correlation $C_z(\boldsymbol{x}_s, \boldsymbol{x}_t)$:** This function captures the correlation between samples s and t arising from their shared z-th stage. We use the standard Gaussian correlation function (squared exponential kernel):

$$C_z(\boldsymbol{x}_s, \boldsymbol{x}_t) = \exp\left(-\sum_{j=1}^{p} \theta_j^{(z)} (x_{js}^{(z)} - x_{jt}^{(z)})^2\right)$$

where $\theta_j^{(z)}$ is the length-scale parameter for the j-th factor in the z-th stage.

- **Stage-wise Interaction Correlation $C_{uv}(\boldsymbol{x}_s, \boldsymbol{x}_t)$:** This function models the correlation contributed by the interaction between the shared u-th and v-th stages. We define it as the product of the corresponding main effect correlations:

$$C_{uv}(\boldsymbol{x}_s, \boldsymbol{x}_t) = C_u(\boldsymbol{x}_s, \boldsymbol{x}_t) \cdot C_v(\boldsymbol{x}_s, \boldsymbol{x}_t)$$

2.2 Step 2: Attention Weights via Softmax

To combine the base functions while ensuring $\Psi_{ss} = 1$, we introduce a set of weights that sum to one. These weights are generated using a Softmax function, representing the model's "attention" to different effects.

- Main effect logits: o_z for $z = 1, \dots, Z$
- Interaction effect logits: o_{uv} for $1 \leq u < v \leq Z$

$$\mathcal{Z} = \sum_{z=1}^{Z} e^{o_z} + \sum_{1 \leq u < v \leq Z} e^{o_{uv}}$$

The corresponding attention weights are then calculated as:

$$w_z = \frac{e^{o_z}}{\mathcal{Z}}, \quad \text{and} \quad w_{uv} = \frac{e^{o_{uv}}}{\mathcal{Z}}$$

By construction, these weights are positive and sum to unity: $\sum_z w_z + \sum_{u<v} w_{uv} = 1$.

2.3 Step 3: The Complete ALLADER Correlation Function

Finally, the correlation Ψ_{st} between any two samples s and t is the weighted sum of all base correlation functions corresponding to their shared stages.

$$\Psi_{st} = \underbrace{\sum_{z=1}^{\min\{z_s, z_t\}} w_z \cdot C_z(\boldsymbol{x}_s, \boldsymbol{x}_t)}_{\text{Weighted Main Effects}} + \underbrace{\sum_{1 \leq u < v \leq \min\{z_s, z_t\}} w_{uv} \cdot C_{uv}(\boldsymbol{x}_s, \boldsymbol{x}_t)}_{\text{Weighted Interaction Effects}}$$

where z_s and z_t are the number of stages for samples s and t, respectively.

3 Key Features and Advantages

1. **Decoupled Design:** The model cleanly separates the global process variance σ^2 from the correlation structure $\mathbf{\Psi}$. This enhances parameter identifiability and interpretability.

2. **Interpretability via Attention:** The learned weights (w_z, w_{uv}) provide direct insight into the importance of each stage (main effect) and each pair of stages (interaction effect) in explaining the data's correlation structure.

3. **Guaranteed Validity:** The sum-to-one constraint on the attention weights ensures that for any sample s, $\Psi_{ss} = 1$, guaranteeing that $\mathbf{\Psi}$ is a valid correlation matrix as required by the GP framework.